

Repurposing of abandoned oil and gas wells as geothermal power plants: A comprehensive sensitivity analysis and AI based performance prediction

Yasin Ahmadpoor, Mozhdeh Sajjadi ^{*}, Mohammad Emami Niri

Institute of Petroleum Engineering, School of Chemical Engineering, College of Engineering, University of Tehran, Iran

ARTICLE INFO

Keywords:

Abandoned wells
Feasibility study
Bottom-hole heat exchanger
Machine learning
Well screening

ABSTRACT

Geothermal energy is a renewable source of energy with immense untapped potentials. The high upfront cost, has made geothermal development projects, economically viable only for high temperature reservoirs. Repurposing of abandoned oil and gas wells as heat exchangers, relieves the drilling cost and presents a promising path for more affordable geothermal energy development. In this manuscript, the energy conservation in a double pipe heat exchanger, has been numerically solved using OpenGeosys, to predict the geothermal energy production from an oil well. The simulation results, under different parameter combinations have been used to rank the involved features, based on a novel perspective for two sets of scenarios with weak and efficient inner pipe insulation. At the end, different machine learning algorithms, have been trained to predict the output temperature of the well. The results showed that the features influencing the performance indicators, depend on the inner-pipe insulation and the output of interest. According to our analyses, in scenarios with weak insulation, the choice of the working fluid and the operational parameters are more important than the screening factors. Among the used ML algorithms, PCA-enhanced SVR demonstrates the best accuracy, with an R^2 score of 0.96 and an RMSE of 1.88.

1. Introduction

As the world's population continues to grow, there will be a simultaneous increase in both the demand and the consumption of energy. In this perspective the future of global energy requirements presents significant challenges. Solutions must focus on sustainable power generation while addressing pressing environmental concerns [1]. Geothermal energy is an abundant, versatile, low-carbon, and sustainable renewable energy resource [2,3]. It has been researched and developed for several decades with significant investment made by governments, private companies and research institutions worldwide. Since 2000, installed geothermal electricity capacity increased at an average annual rate of about 3 % leading to a total installed capacity of approximately 15.96 GWe in 2021. Despite its growth, geothermal energy accounted for a mere 0.5 % of the global renewable electricity market in 2022, while it has the potential to contribute to approximately 2–3 % of global power generation by 2050 (about 800–1300 TWh/yr) [4,5].

Geothermal systems are divided into two main categories: open-loop systems and closed-loop systems. In open-loop systems, such as hydrothermal systems, coproduction systems, and Enhanced Geothermal

Systems (EGS), the circulated fluid is in direct contact with the underground hot rock. Typically, multiple wells are utilized, with at least one well extracting the fluid (usually water) for heat mining and another well returning the fluid to the underground reservoir. Different aspects of open-loop geothermal systems have been extensively studied in the literature [6,7]. Unlike in open-loop systems, where the working fluid flows through the underground formation, in closed-loop systems the fluid is circulated through pipes, being in indirect contact with earth's heat at depth. The closed systems are further classified into two categories: ground source heat pump systems [8,9] and deep borehole heat exchanger (DBHE) systems [10]. Ground source heat pump (GSHP) systems harness shallow geothermal resources for heating and cooling of buildings [11,12], while DBHE geothermal systems can access higher temperatures at depth and be utilized for power generation purposes [10].

The cost of drilling is a significant limitation in the development of geothermal systems, accounting for approximately 50 % of the total project cost [13]. To address this challenge, leveraging the vast number of abandoned oil and gas wells, estimated at 2.5 million wells in the USA and 20–30 million wells globally, presents promising opportunities for

^{*} Corresponding author.

E-mail address: Sajjadi.mozhdeh@ut.ac.ir (M. Sajjadi).

<https://doi.org/10.1016/j.renene.2025.123510>

Received 11 January 2025; Received in revised form 21 March 2025; Accepted 18 May 2025

Available online 21 May 2025

0960-1481/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

geothermal development, leading to cost reduction and shorter development times [14]. A notable achievement in the field of geothermal development occurred in 2021 when MS Energy Solutions Ltd. successfully installed the first closed-loop geothermal system in an abandoned oil well in Kiskunhalas, Hungary, and generated 0.5 MW of heat [15]. Another study on utilization of an oil well in China as a DBHE for heating of a building confirmed the cost-effectiveness of this method compared to conventional heating methods through numerical analyses [16]. Promising pilot projects are also underway in the USA [17,18] and Slovenia [19], currently in advanced stages of implementation. Meanwhile, Organic Rankine Cycle power plants have demonstrated their effectiveness in efficiently generating electricity from working fluid at the temperature range obtained from harnessing abandoned wells as geothermal resources [20,21]. In the period of 2016–2020, the capacity of installed ORC plants in geothermal application showed 45 % increase (+972 MW) with 74 new plants [22].

Despite the promising opportunity abandoned wells propose for the reduction of cost of geothermal energy extraction, there are technical limitations involved with repurposing of oil wells as DBHE systems. The performance and economic viability of the energy extraction process depends on many parameters, most of which are not accurately assessed for oil and gas wells. For instance the thermal properties of the formation rock and the well structure are not of interest during the oil and gas production process and may not be measured at the time of development planning. Overall, the presence and quality of the geothermal resources in the abandoned wells need to be evaluated. Additionally the design parameters such as the flow rate, the size of the well and the inner pipe, and the quality of the insulation between the inner pipe and the well affect the energy harvesting rate. Evaluating the economic viability of repurposing plans, requires careful examination of the energy extraction process and the associated uncertainties.

Various aspects of geothermal energy extraction from deep wells have been investigated in the studies. The numerical modeling of DBHE in vertical oil and gas wells by Kujawa et al., in 2005 was one of the first studies on such systems. They examined the effect of insulation efficiency and inlet water temperature, as design parameters on the energy extraction process [23]. Based on Kujawa's work, numerous studies have directed their attention towards the utilization of abandoned oil wells for geothermal power generation. The models have been improved over the past years [24,25] considering variations in casing geometry [26], and development of multilateral and horizontal wells as DBHE systems [27].

Some of the studies implemented real well data in their modeling. For instance Caulk and Tomac investigated the closed wells in California for converting them into geothermal systems. They utilized a numerical model to examine the temperature of the water exiting the well with respect to different flow rates, depths, and geothermal gradients [28]. Noorollahi et al. numerically modeled two oil wells located in southern Iran in Ahvaz province and looked at the effect of variation of injection rate on the output temperature [26]. The economic and environmental aspects of DBHEs, of a closed oil well in Alberta, Canada was studied by Ali utilizing a small-scale Organic Rankine cycle unit working with water outlet temperature range of 77 °C – 122 °C for electricity generation [29].

The impact of thermal and mechanical properties of the circulated fluid has been the topic of several studies as well. Sui et al. [30] conducted a thorough study on the effects of fluid properties on the performance of geothermal systems. Their study revealed that fluid viscosity, specific heat capacity, and thermal conductivity have significant impact on the energy production while the fluid density does not play an important role. Zhang et al. [31] conducted a comprehensive analysis of nine different working fluids for DBHE systems, including water, CO₂, R134a, R152a, R227ea, R245fa, 1234ze, R600a, and Pentane. In their findings on heat extraction performance, CO₂ demonstrated the highest thermal power generation, with water following closely behind. Isobutane has been considered as one of the

alternatives to water to due to its better thermodynamic properties as well [32,33].

In addition to modeling and experimental studies, the geothermal industry has witnessed a consistent embrace of machine learning techniques over the past 20 years, with a notable exponential rise in applications from 2018 to 2021 [34,35]. Machine learning and deep learning methods have been used for performance prediction of ground source heat pumps (GSHPs) as well [52,53]. Lu et al. [36] used the two-year period hourly data through a field study on a GSHP system to predict different system performance factors. Their study showed that Random Forest model outperform Backpropagation neural network in terms of robustness and average accuracy for prediction. Wardoyo et al. developed an artificial neural network model for one of the wells in the Patuha geothermal field in Indonesia [37]. Jery et al. modeled flow and heat transfer in DBHE and proposed machine learning algorithms for predicting Nusselt number and entropy generation based on numerical results [38].

While extensive research has focused on conventional and enhanced geothermal systems, the utilization of abandoned oil and gas wells for geothermal energy extraction remains a relatively novel technology with limited attention. Similar to other subsurface processes, accurate modeling and analysis are essential due to uncertainties in property assessment and the high cost of development plans. In this manuscript, we perform numerical simulations of a coaxial double pipe DBHE using OpenGeoSys. The simulation data are used for feature ranking through three different algorithms. Additionally, machine learning techniques are proposed to predict output temperature, generated power, and system efficiency as functions of key features. This study aims to enhance our understanding of DBHE systems and evaluate the relative importance of involved parameters.

1.1. Physical model

Among various configurations of DBHE systems, including single U-tube, double U-tube, spiral tube, coaxial double pipe, and helical designs, the coaxial double pipe design stands out due to its superior heat transfer efficiency and ease of installation. Depending on the flow direction, coaxial DBHE systems are categorized into two types: CXC (injecting fluid into the central tube) and CXA (injecting fluid into the annulus) systems. Notably, CXA systems have been reported to exhibit better heat transfer efficiency between the circulating fluid and the surrounding rock formation [39].

Fig. 1 illustrates the geometric parameters involved with the modeling of DBHE systems. The fluid is injected into the annular space and circulated to the surface through the inner tube. The thermal contact between the formation and the fluid, and the absorbance of heat from the subsurface, happens via thermal conduction through the outer pipe wall and the cement behind it. As the heated fluid enters the inner pipe, it is still in thermal interaction with the fluid in the annulus through conduction across the pipe wall. Hence some of the fluid's heat is lost on the way to the surface. Produced hot fluid at the surface is fed into a thermodynamic cycle to transform the energy to electricity. Cool water is returned to the heat exchanger to re-extract heat from the ground.

For the numerical modeling of this process, OpenGeoSys software has been employed. This software is an open-source scientific initiative specifically designed for conducting numerical simulations of thermo-hydro-mechanical-chemical processes in porous media using Finite Element Method [40,41]. Validation of the developed model has been conducted by simulating the experiment of Beier et al. and matching the measured temperature profile of the circulated fluid throughout its path in the well [42]. In their experiments they examined heating of the surrounding rock through circulation of a warm fluid in a double pipe heat exchanger. The physics of the process in their experiment was exactly the same as DBHE systems with reverse heat flow direction. Detailed information of their experiment is provided in Ref. [42]. To reproduce the experiment's conditions in the numerical model, the

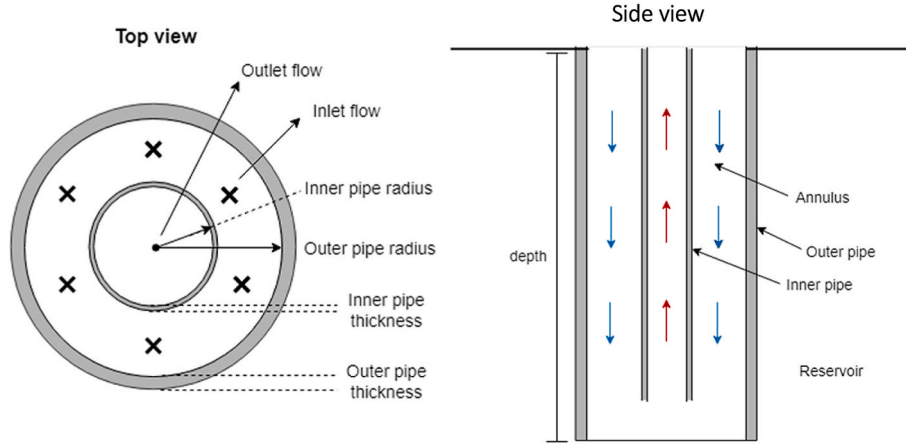


Fig. 1. Schematic of DCBHE with annular inlet.

fluid's temperature profile at the inlet was extracted from the reported graphs in their paper. As Fig. 2a and b shows the fluid temperature results from the numerical simulation exhibit agreement with the experimental measurements.

1.2. Design of numerical experiments

Different simulation scenarios have been designed by choosing 16 properties (listed in Table 1) as variable parameters based on the physics of the model and the reported results of previous sensitivity analyses. For a perceptive analysis, we have categorized the input parameters into Screening Factors, Installation Parameters, Fluid Properties, and Operation Design Parameters. The direct output of the model is the outlet temperature on the surface, at the end of 100 days of operation, which has been recorded for each scenario to evaluate the system's energy production potential at a relatively stabilized stage of operation.

Though the outlet temperature is a mean of evaluation of the system, the gross generated thermal power at the surface Q_{DBHE} is a more practical measure of the performance of the process. The gross generated power depends on the flow rate, q , fluid density, ρ , the specific heat capacity of the fluid, c_p , as well as the increase in the fluid's temperature in the heat exchanger:

$$Q_{DBHE} = c_p \rho q (T_o - T_i) \quad (1)$$

For economic appraisal of energy production units, the analysis is usually based on the coefficient of performance (COP), the ratio of the equivalent produced electricity, E_{DBHE} , to the consumed electricity for running the plant, W_p . Hence the coefficient of performance of the DBHE has been used as another parameter for evaluation of the system:

$$COP = \frac{E_{DBHE}}{W_p} \quad (2)$$

The gained power depends on η_H , the efficiency of the electricity production unit, $E_{DBHE} = Q_{DBHE} \eta_H$. For the oil well bottom hole temperature range of 70–100 °C, Organic Rankin Cycles have been proposed for electric power generation with an efficiency range of 4–18 % [43]. For simplicity, an average efficiency of $\eta_H = 10\%$ has been considered for calculation of COP. W_p in Eq. (2) represents the electric power used for pumping the working fluid into the well and out to the surface and is calculated as [28]:

$$W_p = \frac{q \rho g h_f}{\eta_p} \quad (3)$$

In this equation g is the gravity constant, h_f is the total head loss in the annulus and the inner pipe, and η_p is the pump's efficiency. Head loss in the flow path can be calculated from:

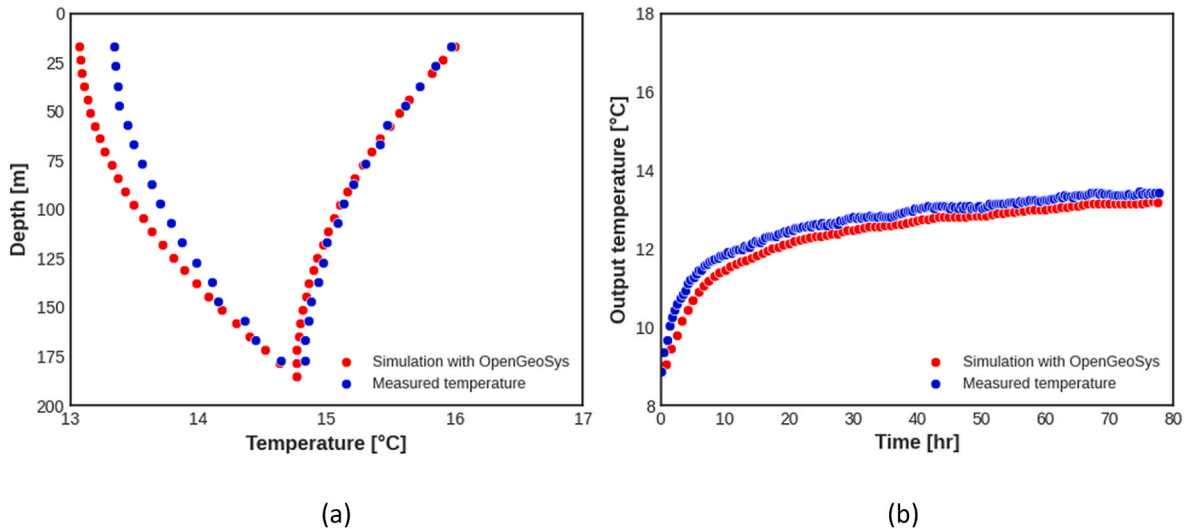


Fig. 2. Comparison between the simulation results versus the measured results by Ref. [47]; a) temperature profile along the flow path from the inlet at the top, to the bottom of the well and then returning through the inner tube to the outlet, b) measured and simulated outlet temperature temporal profiles.

Table 1
List of variables for datasets 1 and 2.

	Variables	mean	standard deviation
Screening Factors	depth [m]	4000	700
	outer pipe diameter	0.14605	0.0105
	formation density [kg/m ³]	2250	250
	formation specific heat capacity [J/kg K]	700	66
	formation thermal conductivity [W/m K]	3.25	0.416
	geothermal gradient [K/m]	0.0325	0.0028
Installation parameters	inner pipe diameter	0.07	0.0146
	inner pipe thickness [m]	0.00937	0.0021
	outer pipe thickness	0.0137	0.0028
	inner pipe thermal conductivity [W/m K]	0.045	0.011 for dataset 1
		22.5	7.5 for dataset 2
Fluid Properties	fluid density [kg/m ³]	1200	133
	fluid viscosity [kg/m s]	0.035	0.0083
	fluid thermal conductivity [W/m K]	2.65	0.783
	fluid specific heat capacity [J/kg K]	3500	666
Operation Design Parameters	Injection flow rate [m ³ /s]	0.00515	0.0016
	inlet fluid temperature [K]	303	10

$$h_f = f_D \frac{LV^2}{2Dg} \quad (4)$$

Where f_D is the friction factor defined using the Reynolds number and relative pipe roughness, ε/D , (ε is 0.025 for steel) [44], and D is the pipe's hydraulic diameter. $\eta_p = 0.4$ has been used as a typical value for pump efficiency.

The defined parameters, namely outlet water temperature, gross heat production, and the coefficient of performance COP of the DBHE system have been used as the target parameters to be predicted by the machine learning algorithms for each scenario. Though these parameters are inherently correlated, each could serve as a different decision making criterion for plant design and economic assessments.

1.3. Feature selection and data preparation

The first step in developing an AI algorithm is to examine the listed features, and if possible, limit the number of involved properties in the prediction process. In this feature selection process, we are looking for the ten most important parameters influencing the efficiency of the system. As stated earlier we are going to look at the parameters in four classes: 1) The formation and well properties, that may be used as screening factors for choosing the right well for the plant implementation, 2) Installation parameters, that may be optimized when designing the system, 3) Fluid properties, that can be used for choosing the suitable agent fluid, and 4) the operation design parameters, that can be

optimized based on the well's performance while the process is running. Beside ranking the features, which will be discussed shortly, categorization of the features can be used as another guideline that shows which stage of the process is more important for assuring an efficient energy production plan.

The inner pipe insulation is one of the installation parameters that has significant influence on the initial investment costs, and is technically challenging to install. It is interesting to examine the relative importance of features, in addition to comparing the system's performance, in presence and absence of the inner pipe insulation, to know if the insulation changes the important features affecting the outlet temperature.

Three datasets have been prepared for this study, two of which were prepared for feature selection. The first data set has been produced for 50 scenarios in systems with efficient insulation, with the thermal conductivity of the insulation layer (inner pipe thermal conductivity) varying between 0.011 and 0.045 W/mK. In the second data set of 100 scenarios, the range of variation of thermal conductivity of the insulation layer has been set to significantly higher values (7.5–22.5 W/mK) to represent systems with very low insulation. In this paper, these two categories of scenarios are plainly referred to as insulated and non-insulated scenarios. The rest of the variables along with their means and standard deviations are similar for the two data sets (Table 1). Although the number of scenarios is limited, due to computational resource constraints in the data generation process, an effort has been made to create data with comprehensive and adequate coverage for the

problem. Each dataset has been generated using the Latin Hypercube Sampling method [45]. The feature importance analysis has been applied to both data sets and the results are compared shortly.

To select the important variables, three methods, namely Pearson's correlation, linear regression feature importance, and extreme gradient boosting (XGBoost) Regressor feature ranking methods have been used.

- Pearson's correlation coefficient quantifies the strength (and the direction) of the linear relationship between each feature x and the output variable y by dividing their covariance, to their standard deviations:

$$r_{xy} = \frac{\sum_{i=1}^M (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^M (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^M (y_i - \bar{y})^2}} \quad (5)$$

Where x_i and y_i represent the data values, and $\bar{x}$ and $\bar{y}$ are mean values of x and y , respectively. M is the sample size.

- The linear regression model also recognizes the linear relationship between each feature and the target variable in terms of the weighting coefficients. These coefficients indicate each feature's importance in defining the output variable's value.
- XGBoost Regressor feature ranking method is a tree-based model. To determine the importance of each feature, the algorithm is trained by the data, and then a score is calculated that indicates how valuable each feature is in the construction of the boosted decision trees within the model.

Each of the algorithms ranks the variables for each output criterion and every set of data with different insulation effectiveness ($3 \times 3 \times 2 = 18$ rankings to be compared). For brevity of the discussions, the

comparison between different ranking methods has been skipped and an average score has been defined based on a feature's scores in different methods:

$$score_{var} = \left[\sum_i^N \frac{(N+1 - rank_i)}{3N} \right] \times 100 \quad (6)$$

Here $rank_i$ is the feature's rank in ranking method $i \in (\text{Pearson's, Linear Regression, XGB})$, and N is the total number of variables (16 in this case). The numerator assigns a minimum score of 1 and a maximum of N to each feature in each method. The ranking results are similar for COP and output temperature, so only the ranking of features based on power generation and COP as the decision-making criteria for non-insulated and insulated scenarios are presented in Tables 2 and 3 respectively. The cells of the tables are colored in accordance to the feature categories of Table 1 for easier visual recognition of the categories in ranking.

The order of ranked features depends on both the insulation conditions and the output variable of interest. Especially for weakly insulated scenarios, the difference between the order of parameters in each of the columns is more significant. Since different methods provide slightly different rankings, the properties with more solid situation in the importance list are going to be chosen. Inlet fluid temperature, and fluid's thermal properties (thermal conductivity and specific heat capacity) are among the top four important features for both ranking criteria when looking at non-insulated scenarios. From the screening factors, the depth and the geothermal gradient are among the top ten important features common for both categories. In general, compared to insulated scenarios, fluid properties are more important and the screening factors are less important when no effective insulation is being considered.

Table 2
Feature ranking for non-insulated dataset.

Rank based on COP		Rank based on Power Generation	
Attribute	importance score (out of 100)	Attribute	importance score (out of 100)
inlet fluid temperature [K]	100	fluid flow rate [m ³ /s]	98
fluid thermal conductivity [W/m K]	92	fluid specific heat capacity [J/kg K]	94
fluid viscosity [kg/m s]	88	inlet fluid temperature [K]	87
fluid specific heat capacity [J/kg K]	75	fluid thermal conductivity [W/m K]	73
depth [m]	67	depth [m]	71
inner pipe thermal conductivity [W/m K]	56	fluid viscosity [kg/m s]	69
formation thermal conductivity [W/m K]	50	geothermal gradient [K/m]	67
geothermal gradient [K/m]	50	inner pipe thermal conductivity [W/m K]	58
fluid flow rate [m ³ /s]	46	inner pipe diameter [m]	42
inner pipe diameter [m]	42	fluid density [kg/m ³]	42
formation density [kg/m ³]	42	formation density [kg/m ³]	40
outer pipe diameter [m]	35	formation thermal conductivity [W/m K]	27
formation specific heat capacity [J/kg K]	33	outer pipe diameter [m]	23
outer pipe thickness [m]	27	formation specific heat capacity [J/kg K]	23
inner pipe thickness [m]	25	inner pipe thickness [m]	21
fluid density [kg/m ³]	23	outer pipe thickness [m]	17

Table 3
Feature ranking for insulated dataset.

Rank based on COP		Rank based on power generation	
Attribute	importance score (out of 100)	Attribute	importance score (out of 100)
depth [m]	96	depth [m]	94
fluid flow rate [m ³ /s]	90	fluid flow rate [m ³ /s]	94
geothermal gradient [K/m]	88	geothermal gradient [K/m]	71
inlet fluid temperature [K]	83	inlet fluid temperature [K]	65
fluid specific heat capacity [J/kg K]	75	formation density [kg/m ³]	62
inner pipe diameter [m]	48	formation thermal conductivity [W/m K]	58
outer pipe thickness [m]	44	outer pipe thickness [m]	54
formation thermal conductivity [W/m K]	44	inner pipe diameter [m]	54
inner pipe thermal conductivity [W/m K]	42	outer pipe diameter [m]	48
fluid viscosity [kg/m s]	42	inner pipe thermal conductivity [W/m K]	46
fluid thermal conductivity [W/m K]	40	fluid specific heat capacity [J/kg K]	46
inner pipe thickness [m]	35	fluid viscosity [kg/m s]	37
fluid density [kg/m ³]	35	fluid thermal conductivity [W/m K]	35
formation specific heat capacity [J/kg K]	31	formation specific heat capacity [J/kg K]	33
outer pipe diameter [m]	29	fluid density [kg/m ³]	29
formation density [kg/m ³]	29	inner pipe thickness [m]	23

For insulated scenarios, the top four features, namely the depth, the fluid flow rate, the geothermal gradient, and the inlet fluid temperature, are the same and appear in the same order for both COP and power generation. Altogether, when the system is equipped with insulation, its performance is less dependent on the fluid properties and is more related to how well the formation provides the system with geothermal power. Also the outer pipe thickness that affects the efficiency of heat extraction from the surrounding rock appears in the top ten features in insulated scenarios.

To summarize, the two operational parameters are common between the two ranking criteria for all scenarios. The inner pipe's properties are among the common important features regardless of the output and the insulation conditions. It should be clarified that in the model the inner pipe thickness only affects the width of the annulus and not the thermal conductivity of the inner pipe, so its variation is not affecting the results. From the screening factors, the depth of the well, the geothermal gradient, and the formation thermal conductivity are the most occurring parameters in the rankings, though their relative importance changes.

Although the fluid properties are important in non-insulated scenarios, the only fluid properties appearing in the most ten important factors for insulated scenarios (and only for COP) are fluid viscosity and fluid specific heat capacity. Fluid's thermal conductivity follows the top ten important features closely. In the following section a more comprehensive analysis of the effect of the studied features, is provided. Yet to avoid verbosity, ten of the features, summarized in Table 4, have been chosen for a statistical sensitivity analysis and for developing the AI algorithms. The base for choosing these features is the ranking obtained for non-insulated scenarios when COP is used for performance evaluation.

1.4. Sensitivity analysis on data sets 1 and 2

The variations in output power with the input variables shown in Figs. 3 and 4 for uninsulated and insulated scenarios respectively are examined. The lines passing through each graph represents the best linear fit to the data points and could be interpreted as the main trend with which the output changes with variation of the input parameters. According to the definition of the rate of heat extraction (given by Eqn 1), power generation depends on the difference between the inlet and outlet temperatures, the flow rate, and fluid's specific heat capacity. The fact that the difference between inlet and outlet temperatures is taken into account in heat extraction, allows for scenarios with lower output temperature to produce more heat if the inlet temperature is low, or if the specific heat capacity of the fluid is large. Also a higher rate of injection results in higher energy production rate.

To discuss the observed trends and their physical implication, we start with the screening factors, namely the wellbore depth, the geothermal gradient, and the formation's thermal conductivity. Increasing the depth of the well results in an increase in the outlet fluid temperature and consequently to higher power generation. The fitted trend line in Fig. 3, for the cases without insulation, shows a less significant dependency on the depth compared to the datapoints corresponding to the insulated scenarios in Fig. 4. This is because, in the absence of insulation on the inner tube, in the return path the fluid loses some of its heat to the cold fluid in the annulus, so the absence of insulation can neutralize the positive effect of the well depth. A similar effect is observed for the geothermal temperature gradient, which intuitively is expected to be an important feature for screening of wellbores for geothermal purposes. For insulated systems the correlation is significant as expected, yet for non-insulated systems the correlation is weaker. Among the screening factors, the generated power shows the

Table 4
List of variables for dataset 3.

	Variables	mean	standard deviation
Screening Factors	depth [m]	4000	700
	formation thermal conductivity [W/m K]	3.25	0.416
	geothermal gradient [K/m]	0.0325	0.0028
Installation parameters	inner pipe diameter	0.07	0.0146
	inner pipe thermal conductivity [W/m K]	0.045	0.011 for dataset 1
		22.5	7.5 for dataset 2
Fluid Properties	fluid density [kg/m ³]	1200	
	fluid viscosity [kg/m s]	0.035	0.0083
	fluid thermal conductivity [W/m K]	2.65	0.783
	fluid specific heat capacity [J/kg K]	3500	666
Operation Design Parameters	Injection flow rate [m ³ /s]	0.00515	0.0016
	inlet fluid temperature [K]	303	10
Fixed parameters	outer pipe diameter	0.14605	
	formation density [kg/m ³]	2250	
	formation specific heat capacity [J/kg K]	700	
	inner pipe thickness [m]	0.00937	
	outer pipe thickness	0.0137	
	fluid density [kg/m ³]	1200	

weakest dependency on the formation thermal conductivity, and the fitted line has a small slope regardless of the effectiveness of insulation.

For non-insulated scenarios, the installation parameters including inner pipe's diameter and its thermal conductivity, show slightly negative effects. For a constant wellbore diameter and injection rate, increasing the inner pipe diameter results in smaller cross-sectional area and higher fluid velocity in the annulus, thus a smaller downhole temperature. The study of outlet temperature (not shown here) shows that the linear trend of output temperature is insensitive to variation of inner pipe diameter. So it seems like the negative effect of this parameter on outlet temperature is alleviated by the larger volume/contact area in the tube and less heat exchange between the injected and produced fluids, on the way back to the surface. Despite the more or less similar output temperature, we still observe a significantly negative slope for non-insulated data set when power generation is considered. This can be attributed to the additional effect of flow rate which has two controversial effects. On one hand, larger injection rate results in larger mass flow rate and higher power generation, and on the other hand it results in lower output heat gain in the annulus. The negative effect is

exacerbated for arrangements with larger inner tubes. For insulated scenarios, as shown in Fig. 4, the trend is very similar to non-insulated scenarios.

Unlike the inner pipe diameter, the effect of inner pipe's thermal conductivity is different for the two insulation scenarios. For non-insulated scenarios, as mentioned earlier, a higher thermal conductivity leads to more heat loss on the return path and a reduced power generation. For insulated scenarios however, the trend is surprisingly reversed and higher thermal conductivity seems to contribute to overall increase of power generation. It seems that although heat loss is in general against thermal energy production, a small amount of heat exchange between the inlet and the outlet flows, allows the scenarios with lower inlet temperature, to absorb the energy from the rock more efficiently. Nevertheless, it should be noted that the range of extracted heat values for insulated scenarios is about 3 times higher than the non-insulated scenarios and that the enhancement in power generation with increased thermal conductivity is only limited to small values of thermal conductivity and effective insulation of the pipe.

The chosen fluid properties, namely the viscosity, the specific heat

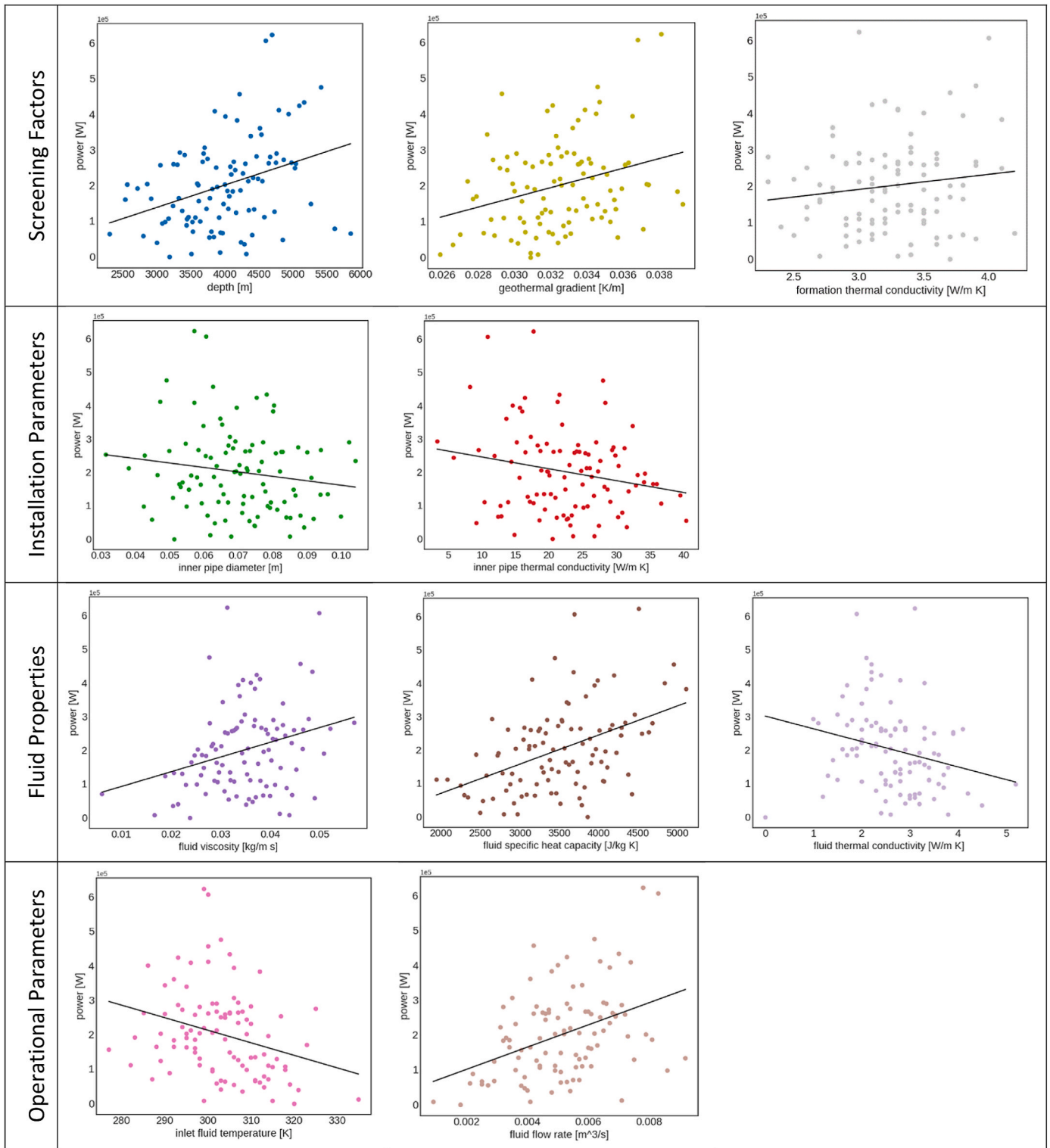


Fig. 3. Influence of important features on the generated thermal power for weakly insulated scenarios.

capacity, and the thermal conductivity, have been ranked higher for non-insulated scenarios and the trend is apparent in Figs. 3 and 4. In the range of studied parameter values, fluid's specific heat capacity has the most significant influence on power generation. For fluids of higher specific heat capacity, more heat is required to raise the temperature of a unit volume of fluid, thus an increase in the specific heat capacity leads to a decrease in the temperature of the fluid at the outlet. In the weakly insulated cases, the reduced heat exchange between the fluid inside the

pipe and the fluid in the annulus due to the smaller temperature difference, results in a slight increase in the outlet fluid temperature, and a considerable positive effect on the power generation rate. The effect is almost negligible for insulated scenarios.

An increase in the thermal conductivity of the fluid results in more efficient transfer of energy from the formation to the heat exchanger, but it also causes the fluid inside the inner pipe to lose its temperature more rapidly during the return to the wellbore. So in insulated cases, where

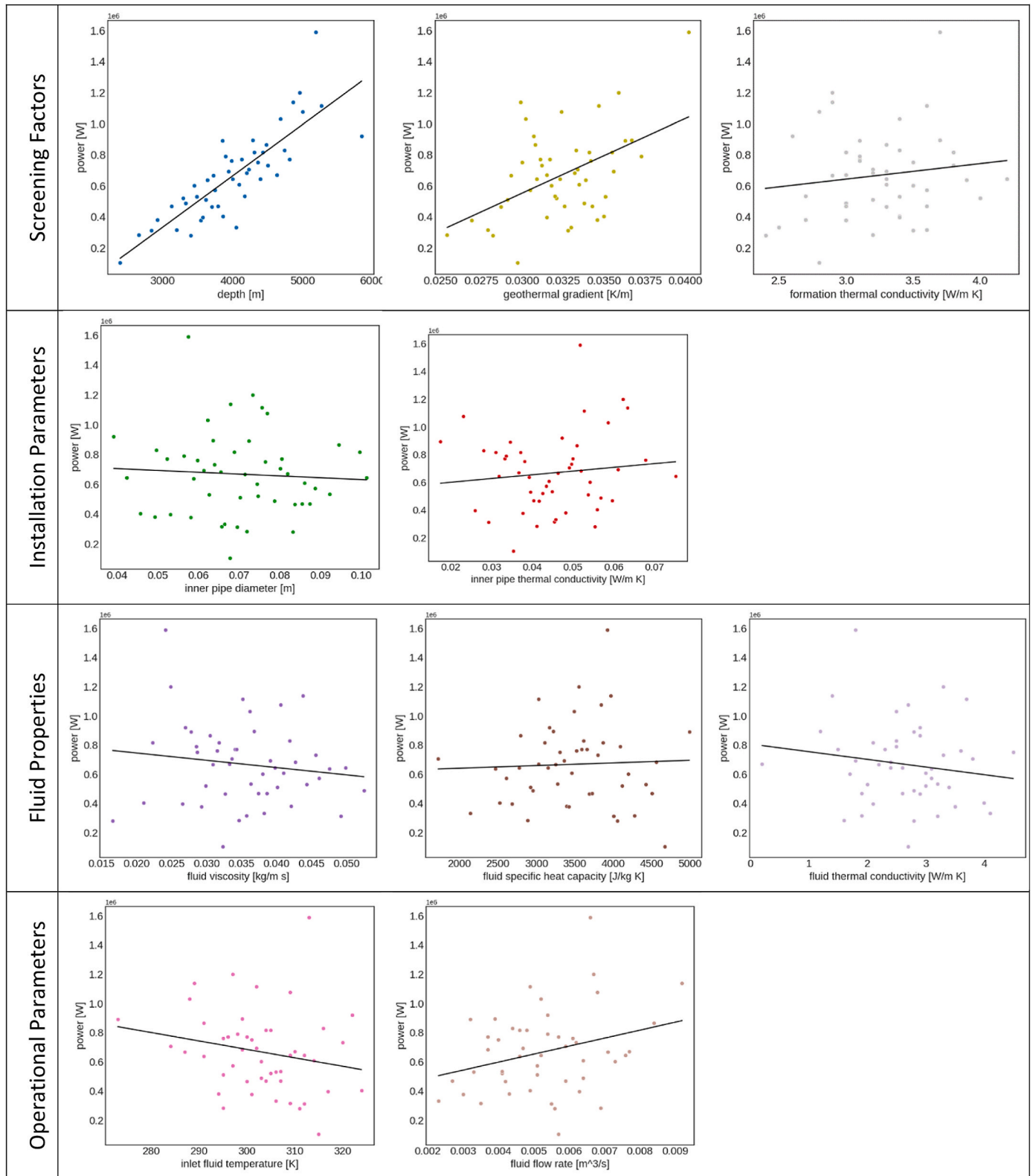


Fig. 4. Influence of important features on the generated thermal power for efficiently insulated scenarios.

the insulation prevents the loss of energy on the return path, with an increase in thermal conductivity, the outlet temperature of the fluid increases but the trend is reversed for non-insulated scenarios. In terms of power generation, for both insulation conditions the output of the system shows a negative effect with the increase of the fluid's thermal conductivity.

An increase in the fluid viscosity results in a reduced convective heat transfer coefficient. Therefore, it is expected, and observed for the insulated scenarios, that the power generation rate (and the outlet fluid temperature) decreases with the increase of the fluid viscosity. However, in the absence of insulation, the outlet fluid temperature (and thus power generation) follows an ascending trend, because a reduction in

the heat transfer coefficient results in less heat loss in the return path to the surface. According to the results it may be concluded that the choice of the proper fluid for the DBHE systems highly depends on presence and absence of insulation.

The design parameters including the injection rate and inlet fluid temperature can be adjusted to optimize the plant's performance, so studying the effect of these parameters is very important. By increasing the injection rate, the duration of contact between the fluid and the

wellbore wall decreases, and results in less heat gain by the fluid from the formation. Consequently, in insulated scenarios, the outlet temperature of the fluid decreases. However, in uninsulated heat exchangers, the faster return of the fluid to the surface means less opportunity to lose its thermal energy inside the inner pipe to the fluid in the annulus and an increase in the outlet temperature. Despite its effect on the outlet temperature, increasing the flow rate has a significant effect on the thermal power generation due to the direct involvement of flow rate in the

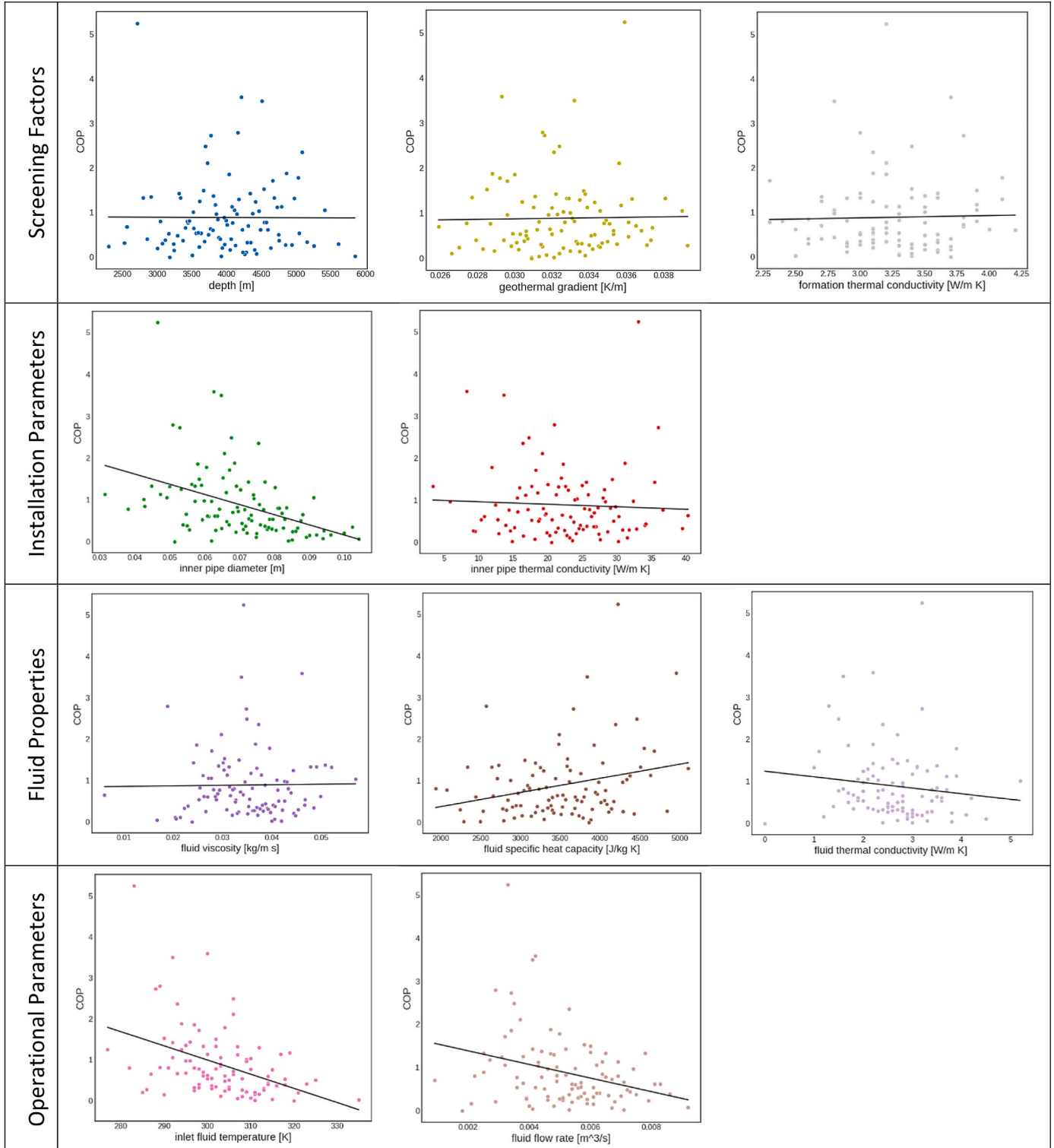


Fig. 5. Influence of important features on coefficient of performance (COP) for weakly insulated scenarios.

corresponding equation. The increase in the inlet temperature, increases the temperature at the outlet, however the heat extraction decreases as the temperature difference between the outlet and the inlet decreases.

The other property to be analyzed is the coefficient of performance, examined for different scenarios versus the parameters of influence in Figs. 5 and 6. This output parameter not only considers the produced thermal energy (converted to the produced electricity), but also accounts for the electric energy consumed for pumping the fluid down the

well and back to the surface. Looking at the screening factors, the effect of such consideration is evident when looking at the effect of the well depth. A deeper well requires higher pumping power weakening the positive effect of higher power generation. So for both insulation conditions the positive slope of power generation vs. depth trendline is relatively alleviated. The negative trends observed for geothermal gradient, and formation thermal conductivity could not be due to their direct effect on COP as they have shown to have positive correlation

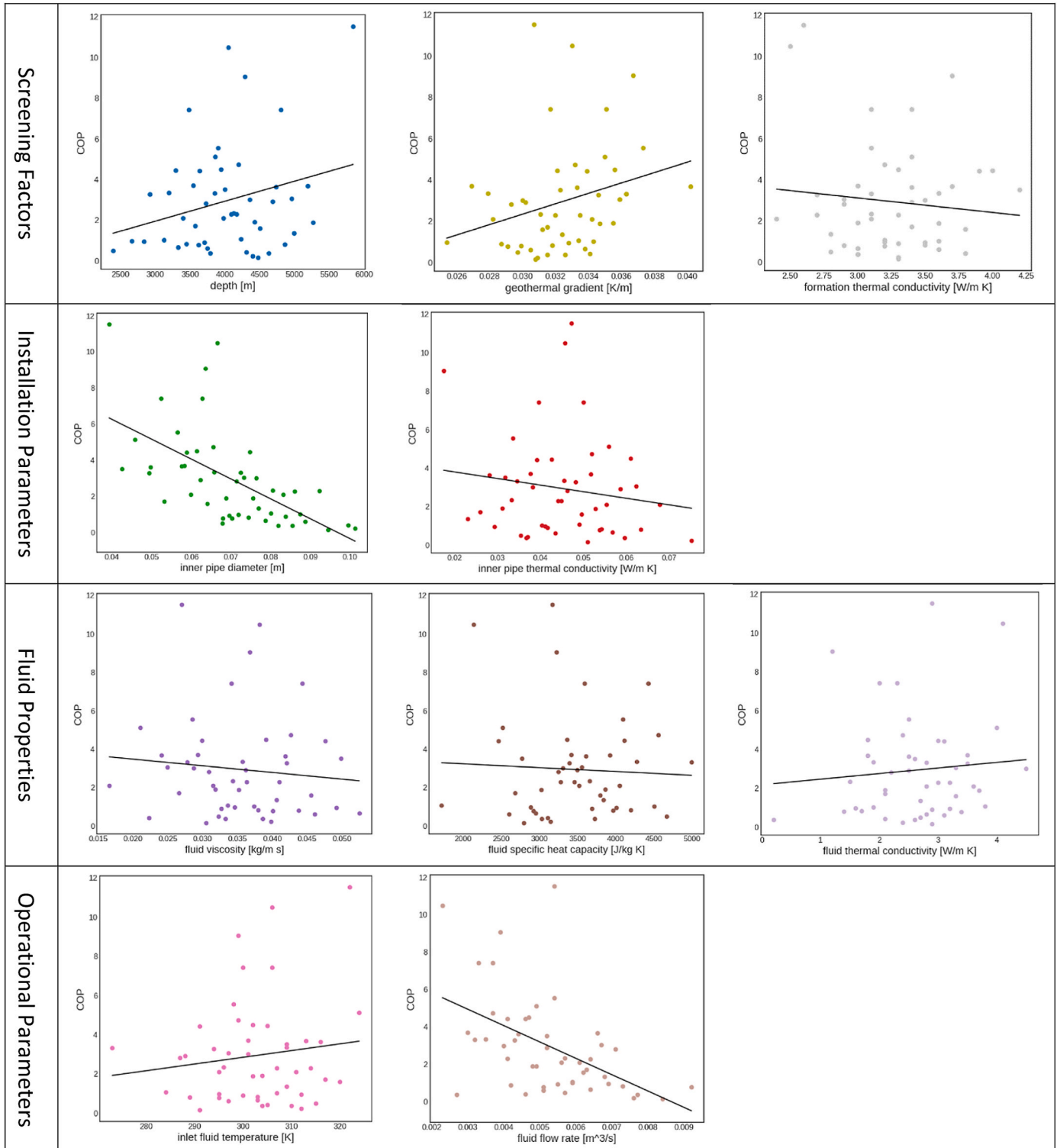


Fig. 6. Influence of important features on coefficient of performance for efficiently insulated scenarios.

with the produced thermal power. However their effect is interrelated to the well depth. Deeper wells amplify the effect of thermal gradient and the positive effect of formation thermal conductivity in the analyses. So when the depth acts against COP, the negative effect is reflected on the dependency of COP on these parameters as well. That is why we see non-significant effect of geothermal gradient on COP for non-insulated scenarios and a comparatively smaller positive trend than what was observed for power generation for insulated scenarios. The power generation rate showed a weak correlation with formation conductivity so the negative impact of the depth is more evidently seen for this parameter that shows a horizontal trendline for non-insulated scenarios and a negative trend for insulated scenarios.

In the category of installation parameters, the increase of inner pipe diameter has a significant negative impact on COP which may be attributed to the increased friction loss in the injection path, in addition to the reduced heat production at the surface. Thermal conductivity of the inner pipe reduces COP but its effect is less pronounced for weekly insulated scenarios. The range of COP values for insulated scenarios is twice that of the COP of weekly insulated scenarios.

Looking at the fluid parameters, a higher fluid viscosity directly results in higher injection pressure. So for weakly insulated scenarios the higher energy consumption negates any positive effect viscosity has on the heat generation rate and for effectively insulated scenarios the negative trend is more pronounced. The effect of fluid's thermal conductivity on COP is similar to the effect observed for output temperature. The fluid's specific heat capacity has a positive effect on COP for non-insulated scenarios but is not much of influence for insulated scenarios.

Among the operational parameters, the injection rate increases power generation (proportional to the injection rate), reduces the output temperature, and increases pumping power (proportional to the square of injection rate) at the same time. The overall effect shows a negative trend for COP versus injection rate for all scenarios. As expected, inlet fluid temperature has a profound effect on the COP as observed for thermal power, for non-insulated scenarios. Application of insulation relieves the dependency of COP on the inlet fluid temperature and even a comparatively positive slope line is observed.

It should be noted that the conducted discussion has been based on linear estimation of the trends, and could be interpreted with precaution. One of the limitations of this method is that the nonlinear dependency of output variables on the examined properties are not touched. Also any correlation between the randomly generated independent variables may affect these trends. The aim of incorporating more complex methods of feature ranking such as XGBoost, is to recognize nonlinear functionality between output variables and different features. So explaining the ranking may not be completely conveyed by the simplified trendline examination. After the feature ranking process and selection of the 10 most important variables based on the COP for non-insulated scenarios, the main dataset consisting of 300 samples has been produced. The remaining non-selected variables have been held constant in the third dataset and fixed at their respective means (Table 4).

1.5. Machine learning models

The remarkable ability of machine learning algorithms to uncover correlations and make accurate predictions has established them as indispensable tools for engineers across various disciplines. This study uses five machine learning algorithms, including Linear Regression, Support Vector Regression (SVR), XGBoost, and two stacking approaches. The first stacking approach involves Linear Regression and XGBoost, and in the second approach, Principal Component Analysis (PCA) and SVR have been incorporated for predictive modeling.

Linear Regression, as a fundamental approach, is utilized for modeling linear relationships between variables. Linear Regression learns from the labeled data and maps the data points to the most

optimized linear functions, which can be used for the prediction on new datasets [46].

SVR is a variant of the Support Vector Machine (SVM) algorithm commonly used for regression tasks. It works by finding a function that best fits the data within a specified margin of error, known as the ϵ margin. Instead of trying to minimize prediction error for every data point, SVR focuses on maintaining a balance between model complexity and prediction accuracy, making it effective for handling both linear and non-linear relationships. In cases where the data is not linearly separable in the original feature space, SVR uses a kernel. In this approach, the data is mapped to a higher-dimensional space where it becomes linearly separable [47,48].

XGBoost is a powerful ensemble method designed for structured data tasks. It collectively combines the predictions of several weak learners, usually decision trees, to create a robust and accurate model. The algorithm is trained iteratively so that each new tree is built to correct the errors of the previous trees, i.e. boosting. This process continues until a certain number of trees are created or a convergence criterion is met [49, 50].

Stacking of algorithms takes an ensemble approach to the next level, blending the strengths of multiple models to enhance predictions. Stacking of Linear Regression and XGBoost involves combining their predictions using a meta-model. Additionally, stacking PCA with SVR allows to leverage PCA's noise reduction capabilities in preprocessing before using SVR as the primary model. PCA works by finding the directions of maximum variance in the data and then projecting the data onto these directions. This can be done by computing the eigenvectors of the data's covariance matrix [51].

1.6. Data preparation for ML models

As mentioned earlier, the main dataset (dataset 3), with 300 scenarios covering the variations of 10 features, has been designed for the prediction of the output variables. These ten variables have been selected according to the feature ranking based on COP for non-insulated scenarios. The remaining 6 least important variables have been fixed to mean values. The input features along with the corresponding outlet temperature at day 100 of the extraction process, are used for training of the ML algorithms and prediction of the outlet temperature. The predicted output temperature can be used for calculation of the other performance evaluation factors examined in the feature ranking section.

1.7. Hyperparameters

The hyperparameters used for SVR and XGBoost, are listed in Table 5. These hyperparameters were chosen by searching through all combinations of the search space, as described in the table, using grid search. The Z-score approach has been used to normalize the data for linear regression, SVR, and PCA-enhanced SVR:

$$z_i = \frac{x_i - \mu}{\sigma} \quad (7)$$

Table 5
Hyperparameters values for SVR and XGBoost Algorithms.

Model	Hyperparameter	Search Space	Selected Value
SVR	C	{0.1, 1, 10, 100}	100
	Gamma	{1, 0.1, 0.01, 0.001}	0.01
XGBoost	Kernel	{rbf, poly, sigmoid}	rbf
	Learning rate	{0.1, 0.01, 0.001}	0.01
	n_estimator	{100, 500, 1000}	1000
	max_depth	{3, 5, 7}	5
	subsample	{0.4, 0.6, 0.8}	0.4
	min_child_weight	{1, 3, 5}	1
	gamma	{0.0, 0.1, 0.2, 0.3}	0.1

Table 6
Performance of models.

Model	Train		Test	
	R ² score	RMSE	R ² score	RMSE
Linear regression	0.94	2	0.83	3.94
SVR	0.97	1.42	0.83	3.95
XGBoost	0.99	0.42	0.90	2.95
Stacking of Linear regression and XGBoost	0.98	1.2	0.93	2.47
PCA-enhanced SVR	0.97	1.42	0.96	1.88

Here x_i represents the parameter value, while μ and σ represent the mean and standard deviation of parameter x , respectively. Furthermore, the number of PCA components is set to 10.

1.8. Performance evaluation metrics

Two metrics, R² and root mean square error (RMSE), described by Eq. (8) and Eq. (9) respectively, have been used to evaluate each algorithm.

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad (8)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^M (y_i - \hat{y}_i)^2}{M}} \quad (9)$$

where $\hat{y}_i$ represents the predicted value of the output y_i . The best values for RMSE and R² vary depending on the type of problem, but generally, a higher R² value closer to one and a lower RMSE value closer to zero indicate improvement in prediction performance.

1.9. Implementation and training

To develop the machine learning models, the dataset is initially divided into training (80 %) and testing (20 %) subsets. Subsequently, the k-fold cross-validation method with 10 splits is applied to the training data, creating training and validation subsets. Next, a grid search technique is employed to explore various hyperparameter combinations to identify the optimal set, based on the RMSE of the validation set. Specifically, grid search is employed for SVR and XGBoost models. Following model training, RMSE and R² metrics are computed

on both the training and testing datasets to assess the overall performance.

2. Results and discussions

This section presents and compares the performance of the named ML algorithms in predicting the outlet temperature based on 300 simulation results for non-insulated scenarios. The values of R² and RMSE for each method applied to training and test data are listed in Table 6. The Linear regression algorithm, achieved an RMSE of 3.94 and an R² score of 0.83. These results indicate that the model has moderate predictive power and can explain approximately 83 % of the variance in the data. This performance shows that the linear regression cannot capture the complex relationships between the outlet temperature and the input variables. The second model, SVR, yielded similar results with an RMSE of 3.95 and an R² score of 0.83. The XGBoost model outperformed the previous two models, providing more accurate predictions and explaining 90 percent of the variance in the data. Fig. 7 shows true and predicted outlet temperature values for the XGBoost model applied on the test data set, and Fig. 8 shows the difference between true and predicted values for the same dataset.

While SVR and XGBoost both have the ability to recognize complex and non-linear relationships, the combination of decision trees in XGBoost seems to better follow the trends of variation of temperature with the given parameters. Since there are both linear and non-linear relationships between the target variable and input variables, a hybrid model of linear regression and XGBoost has been tested in the fourth model. The combination achieved an RMSE of 2.47 and an R² score of 0.93, indicating further improvement in predictive accuracy and explanatory power compared to the individual models. This suggests that the combination of these two models results in a synergistic effect, enhancing the overall performance. Fig. 9 shows true and predicted

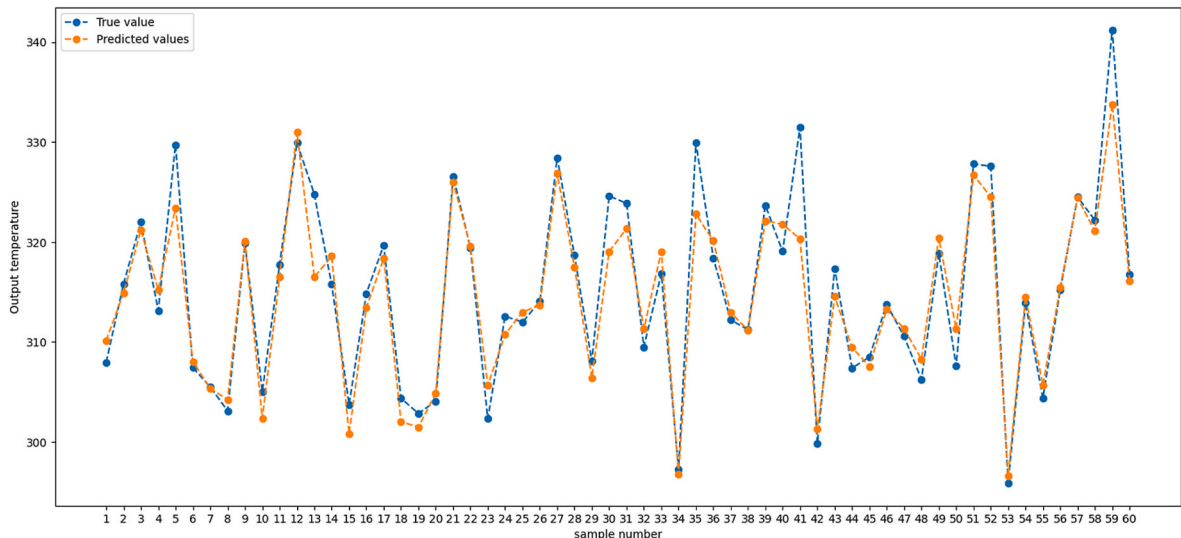


Fig. 7. Comparison of real and predicted outlet temperature of XGBoost.

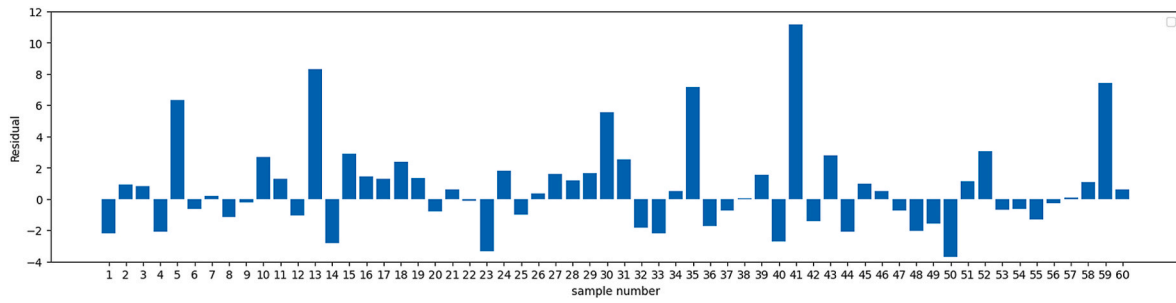


Fig. 8. Residual of real and predicted outlet temperature of XGBoost.

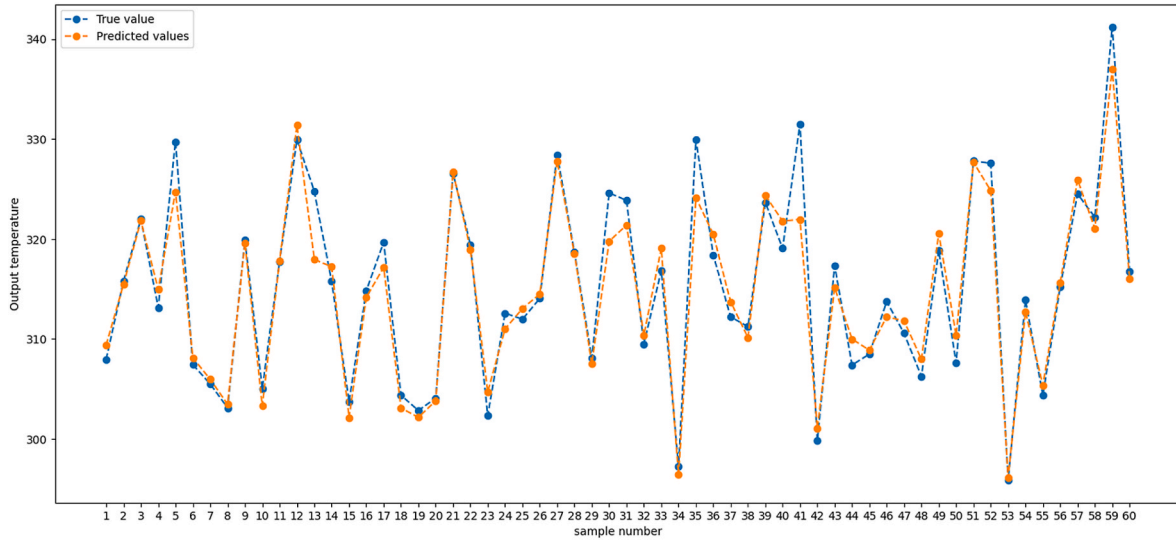


Fig. 9. Comparison of real and predicted outlet temperature of stacking of linear regression and Xgboost.

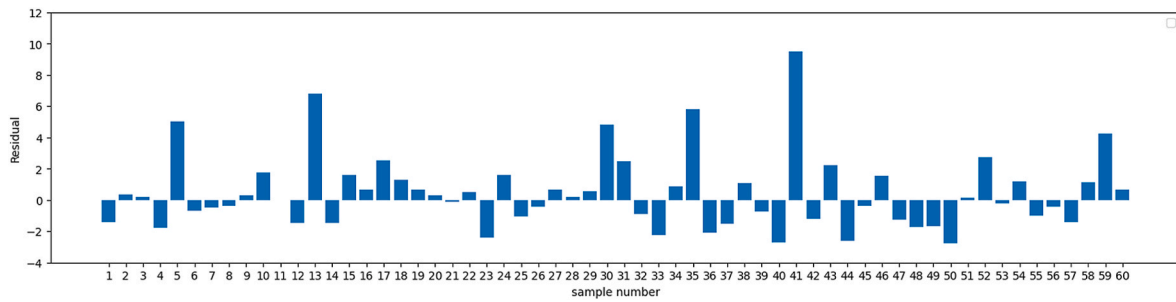


Fig. 10. Residual of real and predicted outlet temperature of stacking of linear regression and Xgboost.

values of outlet temperature for stacking of linear regression and XGBoost model, and Fig. 10 shows the difference between true and predicted outlet temperature for each scenario of the test dataset.

The final model, utilized PCA in conjunction with SVR. Although SVR alone had a weaker performance, when combined with PCA, it exhibited the best performance with the lowest RMSE of 1.88 and the highest R2 score of 0.96. These results suggest that the PCA-enhanced

SVR model provides the best predictive accuracy and explains the highest proportion of variance in the data, making it the most effective model among the options considered. Fig. 11 shows true and predicted values of outlet temperature for the PCA-enhanced SVR model, and Fig. 12 shows the difference between true and predicted outlet temperature for each scenario of the test dataset.

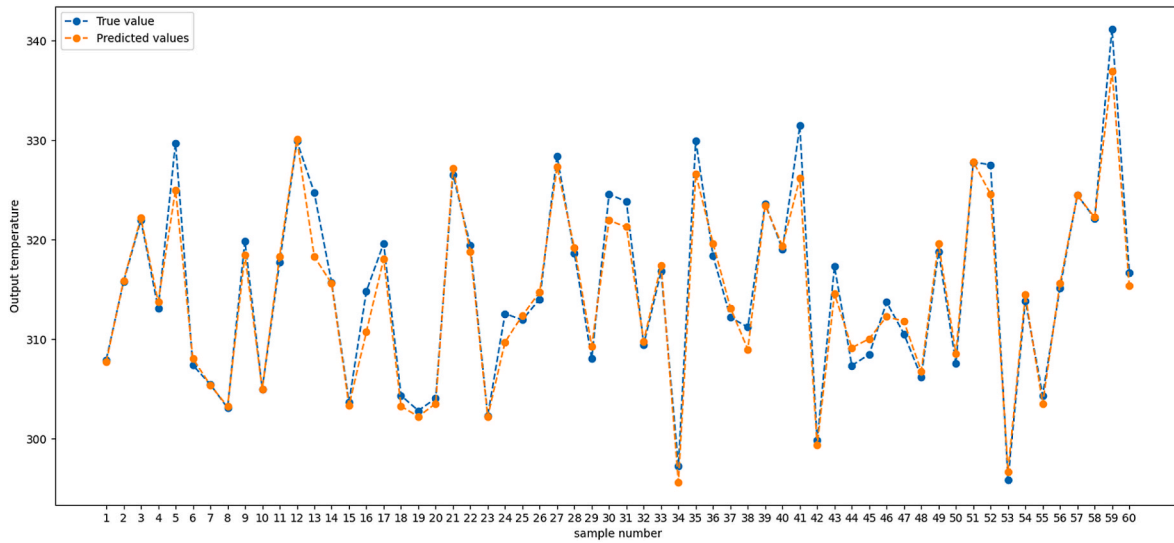


Fig. 11. Comparison of real and predicted outlet temperature of PCA-enhanced SVR.

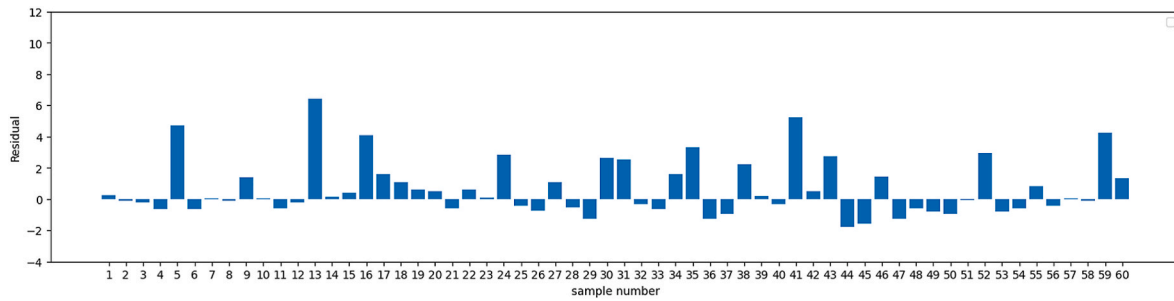


Fig. 12. Residual of real and predicted outlet temperature of stacking of PCA-enhanced SVR.

3. Conclusion

A comprehensive analysis of the relative importance of the features affecting the performance of the DBHE has shown that the lack of insulation on the inner pipe, results in significant loss in energy production capacity from a maximum of 1.6 MW to a Maximum of 0.6 MW. Moreover the fluid properties are more important in weakly insulated systems. So the choice of the right working fluid with high specific heat capacity, low thermal conductivity, and high viscosity is important in such systems. Regardless of the insulation condition, the operational parameters, i.e. the injection rate and the inlet fluid temperature, have significant effect on the performance of the system; though their ranks are higher in insulated scenarios. Additionally, machine learning models were trained on 300 simulations to predict outlet temperature, with PCA-enhanced SVR achieving the best performance ($R^2 = 0.96$, RMSE =

1.88).

CRediT authorship contribution statement

Yasin Ahmadpoor: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation. **Mozhdeh Sajjadi:** Writing – review & editing, Supervision, Investigation, Formal analysis, Conceptualization. **Mohammad Emami Niri:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Nomenclature

c_p	Specific heat capacity of the fluid [J/kg K]	GSHP	Ground source heat pump
f_D	Friction factor	L	Well depth [m]
g	Gravity constant [m/s ²]	M	Number of samples
h_f	Head loss in the annulus and the inner pipe [m]	N	Number of variables
q	Flow rate [m ³ /s]	PCA	Principal Component Analysis
r_{xy}	Pearson's correlation coefficient	Q_{DBHE}	Gross generated thermal power at the surface [W]
x	Input feature vector	RMSE	Root mean square error
x_i	Input feature vector's component	SVM	Support Vector Machine
$\bar{x}$	Input feature's average value	SVR	Support Vector Regressor
y	Target vector	T_i	Inlet fluid temperature [K]
y_i	Target vector's component	T_o	Outlet fluid temperature [K]

(continued on next page)

(continued)

$\bar{y}$	Target vector's average value	V	Fluid velocity [m/s]
$\hat{y}$	Predicted target value	XGBoost	Extreme gradient boosting
z	Z-score	ε	Absolute roughness [m]
COP	Coefficient of performance	η_H	ORC's efficiency (Electric power/Thermal Power)
CXA	The arrangement of DBHE with fluid injected into the annulus	η_p	Pump's efficiency
CXC	The arrangement of DBHE with fluid injected into the central tube	μ	Mean of parameter
D	Pipe's hydraulic diameter [m]	ρ	Fluid density [kg/m ³]
DBHE	Deep borehole heat exchanger	σ	Standard deviation of parameter
E_{DBHE}	Equivalent produced electricity [W]		

References

- [1] C.A. Vargas, L. Caracciolo, P.J. Ball, Geothermal energy as a means to decarbonize the energy mix of megacities, *Commun. Earth Environ.* 3 (2022) 66, <https://doi.org/10.1038/s43247-022-00386-w>.
- [2] N.E. Benti, T.A. Woldegiyorgis, C.A. Geffe, G.S. Gurmesa, M.D. Chaka, Y. S. Mekonnen, Overview of geothermal resources utilization in Ethiopia: potentials, opportunities, and challenges, *Sci. Afr.* 19 (2023) e01562, <https://doi.org/10.1016/j.sciaf.2023.e01562>.
- [3] F.D. Longa, L.P. Nogueira, J. Limberger, J.-D. van Wees, B. van der Zwaan, Scenarios for geothermal energy deployment in Europe, *Energy (Calg.)* 206 (2020) 118060, <https://doi.org/10.1016/j.energy.2020.118060>.
- [4] IRENA and IGA, Global geothermal market and technology assessment, International Renewable Energy Agency (2023). Abu Dhabi; International Geothermal Association, The Hague, <https://www.irena.org/Publications/2023/Feb/Global-geothermal-market-and-technology-assessment>.
- [5] B. van der Zwaan, F.D. Longa, Integrated assessment projections for global geothermal energy use, *Geothermics* 82 (2019) 203–211, <https://doi.org/10.1016/j.geothermics.2019.06.008>.
- [6] L. Santos, A.D. Taleghani, D. Elsworth, Repurposing abandoned Wells for geothermal energy: current status and future prospects, *Renew. Energy* 194 (2022) 1288–1302, <https://doi.org/10.1016/j.renene.2022.05.138>.
- [7] N. Watanabe, G. Blöcher, M. Cacace, S. Held, T. Kohl, Geoenergy Modeling III, Springer, 2017, <https://doi.org/10.1007/978-3-319-46581-4>.
- [8] C.J. Wood, H. Liu, S.B. Riffat, Comparative performance of 'U-tube' and 'coaxial' loop designs for use with a ground source heat pump, *Appl. Therm. Eng.* 37 (2012) 190–195, <https://doi.org/10.1016/j.applthermaleng.2011.11.015>.
- [9] A.M. Omer, Ground-source heat pumps systems and applications, *Renew. Sustain. Energy Rev.* 12 (2008) 344–371, <https://doi.org/10.1016/j.rser.2006.10.003>.
- [10] S. Pan, Y. Kong, C. Chen, Z. Pang, J. Wang, Optimization of the utilization of deep borehole heat exchangers, *Geotherm. Energy* 8 (2020) 6, <https://doi.org/10.1186/s40517-020-0161-4>.
- [11] J. Zhao, Y. Li, J. Wang, A review on heat transfer enhancement of borehole heat exchanger, *Energy Proc.* 104 (2016) 413–418, <https://doi.org/10.1016/j.egypro.2016.12.070>.
- [12] J. Liu, X. Lu, W. Zhang, H. Yu, Numerical investigation of closed-loop heat extraction in different-layout geothermal Wells with particular reference to thermal interference analyses, *Energy (Calg.)* 299 (2024) 131451, <https://doi.org/10.1016/j.energy.2024.131451>.
- [13] E. Barbier, Geothermal energy technology and current status: an overview, *Renew. Sustain. Energy Rev.* 6 (2002) 3–65, [https://doi.org/10.1016/S1364-0321\(02\)00002-3](https://doi.org/10.1016/S1364-0321(02)00002-3).
- [14] S. Kotler, Abandoned oil and gas Wells are leaking. <https://znetwork.org/zmagazine/abandoned-oil-and-gas-wells-are-leaking-by-steven-kotler/>, 2011. (Accessed 21 November 2024).
- [15] A. Richter, First closed-cycle geothermal heat plant set up in Hungary. <https://www.thinkgeoenergy.com/first-closed-cycle-geothermal-heat-plant-set-up-in-hungary/>, 2021. (Accessed 21 November 2024).
- [16] Y.-L. Nian, W.-L. Cheng, Evaluation of geothermal heating from abandoned oil Wells, *Energy (Calg.)* 142 (2018) 592–607, <https://doi.org/10.1016/j.energy.2017.10.062>.
- [17] A. Richter, Call to repurpose abandoned oil Wells in U.S. for geothermal. <http://www.thinkgeoenergy.com/call-to-repurpose-abandoned-oil-wells-in-u-s-for-geothermal/>, 2021. (Accessed 21 November 2024).
- [18] A. Richter, UK geothermal firm launches U.S. business with focus on Texas. <https://www.thinkgeoenergy.com/uk-geothermal-firm-launches-u-s-business-with-focus-on-texas/>, 2021. (Accessed 21 November 2024).
- [19] A. Richter, Plans in motion to utilise geothermal from abandoned oil Wells in Slovenia. <https://www.thinkgeoenergy.com/plans-in-motion-to-utilise-geothermal-from-abandoned-oil-wells-in-slovenia/>, 2021. (Accessed 21 November 2024).
- [20] N.M. Wight, N.S. Bennett, Geothermal energy from abandoned oil and gas Wells using water in combination with a closed wellbore, *Appl. Therm. Eng.* 89 (2015) 908–915, <https://doi.org/10.1016/j.applthermaleng.2015.06.030>.
- [21] M. Macenić, T. Kurevija, Revitalization of abandoned oil and gas Wells for a geothermal heat exploitation by means of closed circulation: case study of the deep dry well Pčelić-1, *Interpretation* 6 (2018) SB1–SB9, <https://doi.org/10.1190/INT-2017-0079.1>.
- [22] C. Wieland, C. Schiffechner, F. Dawo, M. Astolfi, The organic rankine cycle power systems market: recent developments and future perspectives, *Appl. Therm. Eng.* 224 (2023) 119980, <https://doi.org/10.1016/j.applthermaleng.2023.119980>.
- [23] T. Kujawa, W. Nowak, A.A. Stachel, Analysis of the exploitation of existing deep production Wells for acquiring geothermal energy, *J. Eng. Phys. Thermophys.* 78 (2005) 127–135, <https://doi.org/10.1007/s10891-005-0038-1>.
- [24] J.D. Templeton, S.A. Ghoreishi-Madiseh, F. Hassani, M.J. Al-Khawaja, Abandoned petroleum Wells as sustainable sources of geothermal energy, *Energy (Calg.)* 70 (2014) 366–373, <https://doi.org/10.1016/j.energy.2014.04.006>.
- [25] J. Templeton, Abandoned Petroleum Wells as Sustainable/Renewable Sources of Geothermal Energy, Master of Engineering Thesis, McGill University, 2014. <https://escholarship.mcgill.ca/concern/theses/6m311s75w>.
- [26] Y. Noorollahi, M. Pourarshad, S. Jalilinasrabad, H. Yousefi, Numerical simulation of power production from abandoned oil Wells in ahwaz oil field in southern Iran, *Geothermics* 55 (2015) 16–23, <https://doi.org/10.1016/j.geothermics.2015.01.008>.
- [27] G. Wang, X. Song, Y. Shi, F. Yulong, R. Yang, J. Li, Comparison of production characteristics of various coaxial closed-loop geothermal systems, *Energy Convers. Manag.* 225 (2020) 113437, <https://doi.org/10.1016/j.enconman.2020.113437>.
- [28] R.A. Caulk, I. Tomac, Reuse of abandoned oil and gas Wells for geothermal energy production, *Renew. Energy* 112 (2017) 388–397, <https://doi.org/10.1016/j.renene.2017.05.042>.
- [29] B. Ali, Sustainability assessment of power generation from an abandoned oil and gas well in Alberta, Canada, *Journal of Energy and Power Technol* 1 (2019) 2, <https://doi.org/10.21926/jept.1903002>.
- [30] D. Sui, E. Wiktorowski, M. Røksland, T.A. Basmoen, Review and investigations on geothermal energy extraction from abandoned petroleum Wells, *J. Pet. Explor. Prod. Technol.* 9 (2019) 1135–1147, <https://doi.org/10.1007/s13202-018-0535-3>.
- [31] Y. Zhang, C. Yu, G. Li, X. Guo, G. Wang, Y. Shi, C. Peng, Y. Tan, Performance analysis of a downhole coaxial heat exchanger geothermal system with various working fluids, *Appl. Therm. Eng.* 163 (2019) 114317, <https://doi.org/10.1016/j.applthermaleng.2019.114317>.
- [32] W.L. Cheng, T.T. Li, Y.L. Nian, C.L. Wang, Studies on geothermal power generation using abandoned oil Wells, *Energy (Calg.)* 59 (2013) 248–254, <https://doi.org/10.1016/j.energy.2013.07.008>.
- [33] A.P. Davis, E.E. Michaelides, Geothermal power production from abandoned oil Wells, *Energy (Calg.)* 34 (2009) 866–872, <https://doi.org/10.1016/j.energy.2009.03.017>.
- [34] E.R. Okoroafor, C.M. Smith, K.I. Ochie, C.J. Nwosu, H. Gudmundsdottir, M. (J. Aljbran, Machine learning in subsurface geothermal energy: two decades in review, *Geothermics* 102 (2022) 102401, <https://doi.org/10.1016/j.geothermics.2022.102401>.
- [35] L. Wang, Z. Yu, Y. Zhang, P. Yao, Review of machine learning methods applied to enhanced geothermal systems, *Environ. Earth Sci.* 82 (2023) 69, <https://doi.org/10.1007/s12665-023-10749-x>.
- [36] S. Lu, Q. Li, L. Bai, R. Wang, Performance predictions of ground source heat pump system based on random forest and back propagation neural network models, *Energy Convers. Manag.* 197 (2019) 111864, <https://doi.org/10.1016/j.enconman.2019.111864>.
- [37] G.K. Wardoyo, H.B. Pratama, S. Sutopo, A. Ashat, Y. Yudhistira, Application of artificial intelligence in forecasting geothermal production, *IOP Conf. Ser. Earth Environ. Sci.* 732 (2021) 012022, <https://doi.org/10.1088/1755-1315/732/1/012022>.
- [38] A. El Jery, A.K. Khudhair, S.Q. Abbas, A.M. Abed, K.M. Khedher, Numerical simulation and artificial neural network prediction of hydrodynamic and heat transfer in a geothermal heat exchanger to obtain the optimal diameter of tubes with the lowest entropy using water and Al₂O₃/water nanofluid, *Geothermics* 107 (2023) 102605, <https://doi.org/10.1016/j.geothermics.2022.102605>.
- [39] K. Morita, M. Tago, S. Ehara, Case studies on small-scale power generation with the downhole coaxial heat exchanger. <https://api.semanticscholar.org/CorpusID:56387290>, 2005.
- [40] O. Kolditz, S. Bauer, L. Bilke, N. Böttcher, J.O. Delfs, T. Fischer, U.J. Görke, T. Kalbacher, G. Kosakowski, C.I. McDermott, C.H. Park, F. Radu, K. Rink, H. Shao, H.B. Shao, F. Sun, Y.Y. Sun, A.K. Singh, J. Taron, M. Walther, W. Wang, N. Watanabe, Y. Wu, M. Xie, W. Xu, B. Zehner, OpenGeoSys: an open-source initiative for numerical simulation of thermo-hydro-mechanical/chemical (THM/C) processes in porous media, *Environ. Earth Sci.* 67 (2012) 589–599, <https://doi.org/10.1007/s12665-012-1546-x>.
- [41] L. Bilke, T. Fischer, D. Naumov, C. Lehmann, W. Wang, R. Lu, B. Meng, K. Rink, N. Grunwald, J. Buchwald, C. Silbermann, R. Habel, L. Günther, M. Mollaali,

- T. Meisel, J. Randow, S. Einspänner, H. Shao, K. Kurgiyis, O. Kolditz, J. Garibay, OpenGeosys Version 6.4.3, Zenodo, 2022. <https://zenodo.org/records/7092676>.
- [42] R.A. Beier, J. Acuña, P. Mogensen, B. Palm, Transient heat transfer in a coaxial borehole heat exchanger, *Geothermics* 51 (2014) 470–482, <https://doi.org/10.1016/j.geothermics.2014.02.006>.
- [43] X. Liu, G. Falcone, C. Alimonti, A systematic study of harnessing low-temperature geothermal energy from oil and gas reservoirs, *Energy (Calg.)* 142 (2018) 346–355, <https://doi.org/10.1016/j.energy.2017.10.058>.
- [44] V. Bellos, I. Nalbantis, G. Tsakiris, Friction modeling of flood flow simulations, *J. Hydraul. Eng.* 144 (2018) 04018073, [https://doi.org/10.1061/\(ASCE\)HY.1943-7900.0001540](https://doi.org/10.1061/(ASCE)HY.1943-7900.0001540).
- [45] M.D. McKay, R.J. Beckman, W.J. Conover, A comparison of three methods for selecting values of input variables in the analysis of output from a computer code, *Technometrics* 21 (1979) 239–245, <https://doi.org/10.2307/1268522>.
- [46] T. Hastie, R. Tibshirani, J.H. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, vol. 2, Springer, New York, 2009. <https://link.springer.com/book/10.1007/978-0-387-84858-7>.
- [47] C. Cortes, V. Vapnik, Support-vector networks, *Mach. Learn.* 20 (1995) 273–297, <https://doi.org/10.1007/BF00994018>.
- [48] N. Rasifaghghi, *From Theory to Practice: Implementing Support Vector Regression for Predictions in Python*, 2023. (Accessed 21 November 2024).
- [49] T. Chen, C. Guestrin, XGBoost: a scalable tree boosting system, *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (2016) 785–794, <https://doi.org/10.1145/2939672.2939785>.
- [50] R. Guo, Z. Zhao, T. Wang, G. Liu, J. Zhao, D. Gao, Degradation state recognition of piston pump based on ICEEMDAN and XGBoost, *Appl. Sci.* 10 (2020) 6593, <https://doi.org/10.3390/app10186593>.
- [51] I. Jolliffe, *Principal component analysis for special types of data*, in: *Principal Component Analysis*, Springer, New York, 2002, pp. 338–372, https://doi.org/10.1007/0-387-22440-8_13.
- [52] J.-H. Shin, Y.-H. Cho, Machine-learning-based coefficient of performance prediction model for heat pump systems, *Appl. Sci.* 12 (2022) 362, <https://doi.org/10.3390/app12010362>.
- [53] S.K. Park, H.J. Moon, K.C. Min, C. Hwang, S. Kim, Application of a multiple linear regression and an artificial neural network model for the heating performance analysis and hourly prediction of a large-scale ground source heat pump system, *Energy Build.* 165 (2018) 206–215, <https://doi.org/10.1016/j.enbuild.2018.01.029>.